

AI-Powered Math Learning Platform for Sec & JC Students

Overview of our project objectives, team members, and our progress with live demo

• Upload Questions & Practice Papers

• AI-assisted marking & feedback

• Dynamic Question Generation

OBJECTIVE AND MISSION

OBJECTIVE

Empower Singapore Students to
Achieve Academic Excellence Through
AI-Powered Learning

MISSION

Revolutionise the way students learn
by incorporating AI into their learning
experience

01 Practice AI-generated math papers

02 AI-assisted marking & feedback

03 Performance analytics & progress report

04 Personalised study planner

05 Interactive notes with visuals & simulations

TEAM MEMBERS

Four Software Developers

We are a team of full-stack developers in our penultimate year of study, majoring in Data Science and Artificial Intelligence. We are passionate about harnessing AI to enhance students' learning experiences.



Jun Sheng

Cloud Developer & SEP
liaison



Isaac

Full-stack Developer



Kee Qing

Full-stack Developer



Hong Sheng

Full-stack Developer



Key Baseline Features of the app

Focus on Secondary O-Level students, GCE 4049 Additional Mathematics. Built to expand to more subjects and levels.

Add a paper

A past-year exam paper goes in, ready for students to attempt.

Submit Answer Key

Answer keys can be submitted for marking and providing solutions.

Generate More

Question bank contains past questions and generates new ones.

Student attempts

They upload a photo or scan of their handwritten work.

Acerie marks it

Every question is read and marked against the answer key.

Feedback & Planner

The student reviews mistakes and plans their next practice.

What we are building next.

Four priorities take the POC from "marks a paper" to "*guides a student*".

NEXT

Advanced Question & paper generation

Generate fresh practice questions and full papers from the syllabus, beyond past-year stock, utilizing orchestrators and feedback loops.

NEXT

Smarter feedback

Feedback that draws on past corrections and worked solutions, so it stays sharp and consistent.

NEXT

Learning planner

A personalised study schedule that targets weak topics on the way to exam day.

NEXT

Topic breadth & quality

More topics to practise, with validation that generated questions hold up.

acerie

questions | Table Editor | Acerie

Secondary School - Google

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ACERIE

EXAM ATELIER

ACCOUNT

KQing

ADMIN

@CHAN1203

Control Room

Paper Library

Review Grading

Learning

Populate Paper

Students

Sign out

ADMIN STUDIO

Control, curate, create

Populate the question bank

This is the operational surface for curating reference solutions so grade 12 students can answer questions.

PUBLISH PRACTICE PAPER

PAPER TITLE

Sec 4 Algebra Sprint

LEVEL

No level filter

TOPIC

Algebra

SUBTOPIC

Algebraic Manipulation

QUESTION SEARCH

Source Paper

Paper 1

Upload a source paper PDF

question-ingestion pipeline and inserts extracted questions into the bank.

Run ingestion →

ATTACH REFERENCE ANSWER

SOURCE PAPER

BANDICAM

UNREGISTERED

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